DSP – LEC – 10-2 - Ch-4

- Residue

CHAPTER- 4- cont.

Residue method

LEC. 10-2- index

4.3 The inverse Z-transform – **cont.**

4.3.3 Residue method

4.3.4 Comparison of inverse z-tran. methods

4.4 Properties of the Z-Transform

Sec. 4.3.3

4.3.3 Residue method

- In this method the IZT is obtained by evaluating the contour integral

$$x(n) = \frac{1}{2\pi j} \oint_C z^{n-1} X(z)\, dz \qquad (4.23)$$

- where C is the path of integration enclosing all the poles of X(z).

Sec. 4.3.3 Cont. - 1

- For rational polynomials, the contour integral in **Eq. 4.23** is evaluated using a fundamental result in complex variable theory known as Cauchy's residue theorem (Mathews, 1982):

$$x(n) = \frac{1}{2\pi j} \oint_C z^{n-1} X(z)\, dz \qquad (4.24)$$

= sum of the residues of $z^{n-1}X(z)$ at all the poles inside C.

Sec. 4.3.3 Cont. - 2

- In the partial fraction method, it was stated that the partial fraction coefficients, the C_k, are also referred to as residues of **X(z)** and a way of obtaining their values given.

- The key point to remember is that every residue, C_k, is associated with a pole, p_k.

- In the present method, the residue of $z^{n-1}X(z)$ at the pole p_k (not the residue of X(z)) is given by:

$$\text{Res}\,[F(z), p_k] = \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-p_k)\,F(z)\right]_{z=Pk} \qquad (4.25)$$

Sec. 4.3.3 Cont. - 3

$$\text{Res}\,[F(z),\ p_k] = \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-p_k)\,F(z)\right]_{z=p_k} \qquad (4.25)$$

Where: $F(z) = z^{n-1}X(z)$,

m is the order of the pole at p_k and

Res [F(z), p_k] is the residue of **F(z) at z = p_k** .

- For a simple (distinct) pole, **Eq. 4.25** reduces to

$$\begin{aligned}\text{Res}[F(z),\ p_k] &= (z-p_k)\,F(z)\\ &= (z-p_k)z^{n-1}\ X(z)\big|_{z=p_k}\end{aligned} \qquad (4.26)$$

Sec. 4.3.3 Cont. - 4

Example 4.7 **(X(z) contains simple, first-order poles)**

Find, using the **residue method**, the discrete-time signal corresponding to the following z-transform:

$$X(z) = \frac{z}{(z - 0.75)(z + 0.5)}$$

Assume **C** is the circle |z| = 1.

Sec. 4.3.3 Cont. - 5

Solution

- This problem is the same as **Example (4.4)**.
- In factored form, X(z) is given by:

$$X(z) = \frac{z}{(z-0.75)(z+0.5)}$$

- If we let **$F(z) = z^{n-1} X(z)$** then:

$$F(z) = \frac{z^{n-1} z}{(z-0.75)(z+0.5)}$$

$$= \frac{z^{n}}{(z-0.75)(z+0.5)}$$

Sec. 4.3.3 Cont. - 6

- **F(z)** has poles at **z = 0.75 and z=-0.5**.

- Both poles lie inside the unit circle in **Fig. 4.3**.

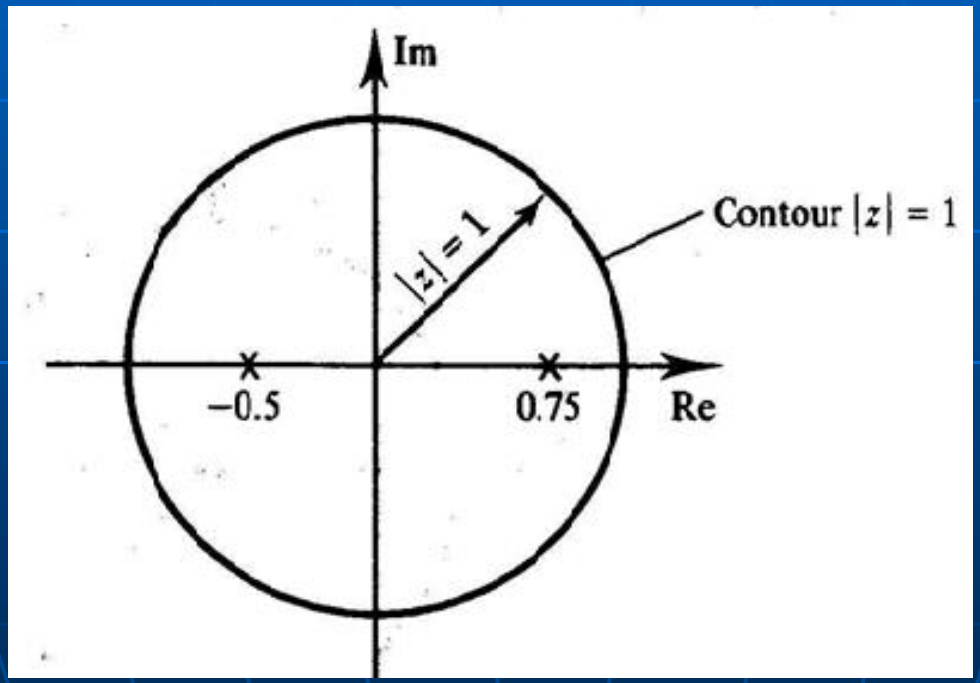


Fig. 4.3 Showing the poles of **X(z)**.

Sec. 4.3.3 Cont. - 7

$$x(n) = \frac{1}{2\pi j}\oint_C z^{n-1} X(z)\, dz$$

$$= \text{sum of the residues of } z^{n-1}X(z) \text{ at all the poles inside C.} \qquad (4.24)$$

- From **Eq. 4.24**, the inverse z-transform is given by:

$$x(n) = \text{Res}\,[F(z), 0.75] + \text{Res}\,[F(z), -0.5]$$

Sec. 4.3.3 Cont. - 8

$$\text{Res}[F(z), p_k] = (z - p_k)\, F(z) = (z - p_k) z^{n-1}\, X(z)\big|_{z=p_k} \qquad (4.26)$$

- Since the poles are first order, **Eq. 4.26** will be used. Thus:

$$\text{Res}[F(z), 0.75] = (z-0.75)\, F(z)\big|_{z=0.75}$$

$$= \left.\frac{(z-0.75)z^n}{(z-0.75)(z+0.5)}\right|_{z=0.75} = \frac{(0.75)^n}{0.75+0.5}$$

$$= \frac{4}{5}(0.75)^n$$

$$\text{Res}[F(z), -0.5] = (z+0.5)\, F(z)\big|_{z=-0.5}$$

$$= \left.\frac{(z+0.5)z^n}{(z-0.75)(z+0.5)}\right|_{z=-0.5} = -\frac{4}{5}(-0.5)^n$$

Sec. 4.3.3 Cont. - 9

- The **IZT** is the sum of the residues at **z = 0.75** and at **z = -0.5**:

$$x(n) = (4/5)[(0.75)^n - (-0.5)^n]$$

which is identical to the result obtained by the partial fraction expansion.

Sec. 4.3.3 Cont. - 12

Example 4.9 (X(z) contains a second-order pole)

Find the discrete-time sequence, x(n), with the following z-transform :

$$X(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

Solution

This example is the same as **Example (4.6)** under partial fraction expansion.

Sec. 4.3.3 Cont. - 13

- According to the residue method the discrete-time sequence is given by:

$$x(n) = \sum_{k=1}^{M} \text{Res}\,[F(z), p_k]$$

- Where:

$$F(z) = z^{n-1}\,X(z) = \frac{z^{n+1}}{(z-0.5)(z-1)^2}$$

Sec. 4.3.3 Cont. - 14

- **F(z)** has **a simple pole at z = 0.5** and **a second-order pole at z = 1;** thus **x(n)** is given by:

$$x(n) = \text{Res}\,[F(z), p_1] + \text{Res}\,[F(z), p_2]$$

$$\text{Res}\,[F(z), 0.5] = \frac{(z-0.5)z^{n+1}}{(z-0.5)(z-1)^2} = \left.\frac{z^{n+1}}{(z-1)^2}\right|_{z=0.5}$$

$$= 0.5(0.5)^n/(0.5)^2$$

$$= 2(0.5)^2$$

$$= (0.5)^{n-1}$$

Sec. 4.3.3 Cont. - 15

$$\text{Res}\,[F(z),\,1] = \frac{d}{dz}\left[\frac{(z-1)^2 z^{n+1}}{(z-0.5)(z-1)^2}\right]$$

$$= \left.\frac{(z-0.5)(n+1)z^n - z^{n+1}}{(z-0.5)^2}\right|_{z=1}$$

$$= [(0.5)(n+1) - 1] / (0.5)^2$$

$$= 2(n-1)$$

- Combining the results, we have:

$$\mathbf{x(n) = 2[(n-1) + (0.5)^n]}$$

which is the same result as for the **partial fraction expansion method**

Sec. 4.3.3 Cont. - 16

- The reader may have noticed that **the partial fraction** and **residue methods** are related.

- Both methods required the evaluation of residues albeit performed in different ways.

- The partial fraction method requires the evaluation of the residues of **X(z)**, i. e. the **C_k, while** the residue method requires the evaluation of the residues **of $z^{n-1}X(z)$.**

- When X(z) has first-order poles we have:

$$\text{Res}[z^{n-1}X(z), p_k] = z^n \text{Res}[X(z), p_k] = z^n C_k \qquad (4.28)$$

Sec. 4.3.4

4.3.4 Comparison of the inverse z-trans. methods

- We have discussed in some detail three methods of obtaining the IZT: the power series, partial fraction expansion and the residue method.

- **A limitation of the power series method is:** that it does not lead to a closed form solution (although this can be deduced in simple cases), but it is simple and lends itself to computer implementation.

Sec. 4.3.4 Cont. - 1

- However, because of its recursive nature care should be taken to minimize possible build-up of numerical errors when the number of data points in the inverse z-transform is large, **e. g.,** by using double precision.

- **Both** the partial fraction expansion and the residue methods lead to **closed form solutions**.

- **The main disadvantage with both methods** is the need **to factorize the denominator polynomial**, **i. e.,** at is finding the poles of X(z).

Sec. 4.3.4 Cont. - 2

- **If the order of X(z)** is high finding the poles of **X(z),** if **X(z)** is not in factored form, is quite a difficult task.

- **Both** methods may also involve high-order differentiation if X(z) contains multiple-order poles.

- **Clearly**, if closed form solution is required then the partial fraction or residue method is the most appropriate.

Sec. 4.3.4 Cont. - 3

- **The partial fraction method is** particularly useful in generating the coefficients of parallel structures for digital filters.

- **The residue method is** widely used in the analysis of quantization errors in discrete-time systems.

4.4 PROPERTIES OF THE Z-TRANSFORM

Some useful properties of the z-transform which have found practical use in DSP are described briefly below.

1- Linearity:

If the sequences $x_1(n)$ and $x_2(n)$ have z-transforms $X_1(z)$ and $X_2(z)$, then the z-transform of their linear combination is

$$ax_1(n) + bx_2(n) \rightarrow aX_1(z) + bX_2(z) \qquad (4.29)$$

Sec. 4.4 Cont. - 1

2- Delays or shifts:

- If the z-transform of a sequence, x(n), is X(z) then the z-transform of the sequence delayed by **m** samples is z – m X(z).

- This property is widely used in converting the z transfer function of discrete-time systems into time domain difference equations and **vice versa.**

$$x(n) \rightarrow X(z)$$

$$x(n-m) \rightarrow z^{-m} X(z)$$

Sec. 4.4 Cont. - 2

3- Convolution:

Given a discrete-time LTI system with input, **x(n),** and impulse response, **h(k)**, the output of the system is given by:

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k) \qquad \textbf{(4.30a)}$$

Sec. 4.4 Cont. - 3

- In terms of the z-transform, the input and output are related as:

$$Y(z) = H(z)\,X(z) \qquad (4.30b)$$

- **Where;** X(z), H(z) and Y(z) are, respectively, the z-transform of x(n), h(k) and y(n).

Given X(z) and H(z), the output y(n) can be obtained by inverse z-transform Y(z).

Sec. 4.4 Cont. - 4

- It is seen that the convolution operation in **Eq. 4-30a** has **become a multiplicative process in the z-domain.**

- **H(z)** is often referred to as the **system transfer function.**

Sec. 4.4 Cont. - 5

4- Differentiation:

- If X(z) is the z-transform of x(n), then the z-transform of **n x(n)** can be obtained by differentiating **X(z):**

$$x(n) \rightarrow X(z)$$

$$n\,x(n) \rightarrow -z\,\frac{dX(z)}{dz} \qquad (4.31)$$

- This property is useful in obtaining the **IZT** when **X(z)** contains multiple order poles.

Sec. 4.4 Cont. - 6

5- Relationship with the Laplace transform.

- Continuous-time systems **or** signals are normally described using the **Laplace transform**.

- If we let $z = e^{sT}$,

where, **s** is the complex Laplace variable given by:

$$s = \sigma + j\omega$$

- **Then**

$$z = e^{sT} \quad e^{(\sigma + j\omega)T} = e^{\sigma T} \; e^{j\omega T} \qquad (4.32)$$

thus

$$|z| = e^{\sigma T} \text{ and } \angle z = \omega T = 2\pi f/F_s = 2\pi\,\omega/\omega_s$$

Sec. 4.4 Cont. - 7

thus

$$|z| = e^{\sigma T} \text{ and } \angle z = \omega T = 2\pi f/F_s = 2\pi \omega/\omega_s$$

where ω_s (rads^{-1}) is the sampling frequency.

- As **ω** varies from **- ∞ to ∞** the **s-plane** is mapped to the **z-plane** as shown in **Fig.4.5**.

- The entire **jω** axis in the **s-plane** is mapped onto the unit circle.

- **The left-hand s-plane** is mapped to the inside of the unit circle and **the right-hand s-plane** maps to the outside of the unit circle.

Sec. 4.4 Cont. - 8

- In terms of frequency response, the **jω** axis is the most important in the **s-plane**.

- In this case, **d = 0**, and frequency points in the **s-plane** are related to points on the **z-plane unit circle** by:

$$Z = e^{j\omega T} \qquad \textbf{(4.33)}$$

Sec. 4.4 Cont. - 9

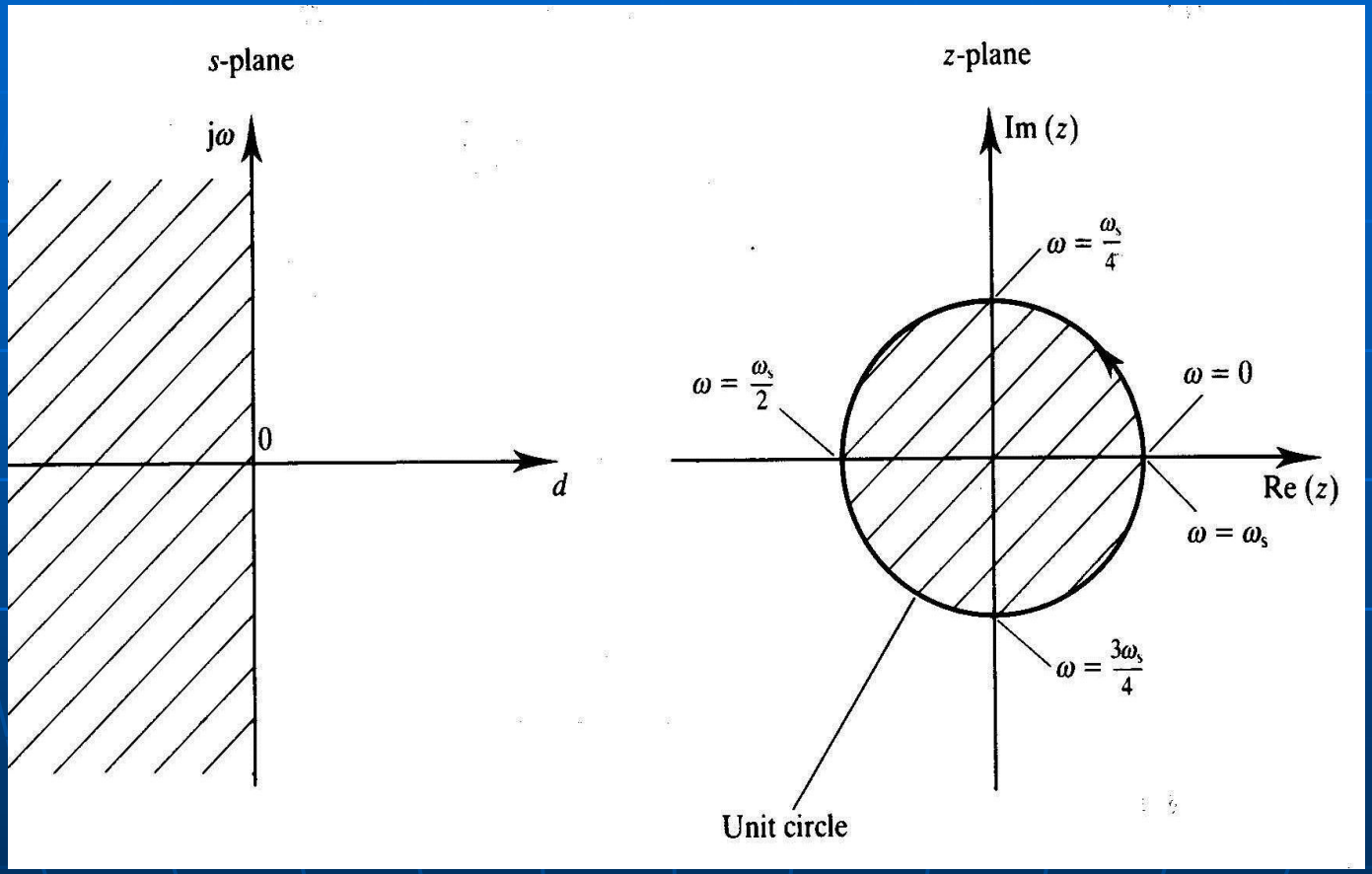


Fig. 4.5 Mapping of **s-plane** to **z-plane**. The left-hand side of the s-plane maps to the inside of the unit circle, the right-hand side maps to the exterior of the unit circle and the j-axis maps onto the unit circle.

Sec. 4.4 Cont. - 10

- **Table 4.2** shows how some specific frequencies are mapped from the **s-plane** to the **z-plane**.

- It is clear that the mapping is not **unique** since for example the two frequencies **$\omega=\omega_s$ and $\omega=2\omega_s$** in the **s-plane** map to the same point on the unit circle.

Sec. 4.4 Cont. - 11

$$|z| = e^{\sigma T} \text{ and } \angle z = \omega T = 2\pi f/F_s = 2\pi \omega/\omega_s$$

Table 4.2 Mapping of frequencies from the s-plane to the z-plane.

s-plane : ω(rads^{-1})	z-plane : ωT (rad)
0	0
$\omega_s/4$	$\pi/2$
$\omega_s/2$	π
$3\omega_s/4$	1.25π
ω_s	2π
$1.25\omega_s$	$\pi/2$
$1.5\omega_s$	π
$1.75\omega_s$	1.25π
$2\omega_s$	2π